

SEQUENTIAL CONSTRUCT

PROBLEM 1

```
.data
a_ap: .word 1
d_ap: .word 10
a_gp: .word 4
r_gp: .word 2

.text
.globl main
main:
    lw  $t0, a_ap
    lw  $t1, d_ap

    move $s0, $t0
    add  $s1, $s0, $t1
    add  $s2, $s1, $t1
    add  $s3, $s2, $t1
    add  $s4, $s3, $t1
    add  $s5, $s4, $t1
    add  $s6, $s5, $t1
    add  $s7, $s6, $t1

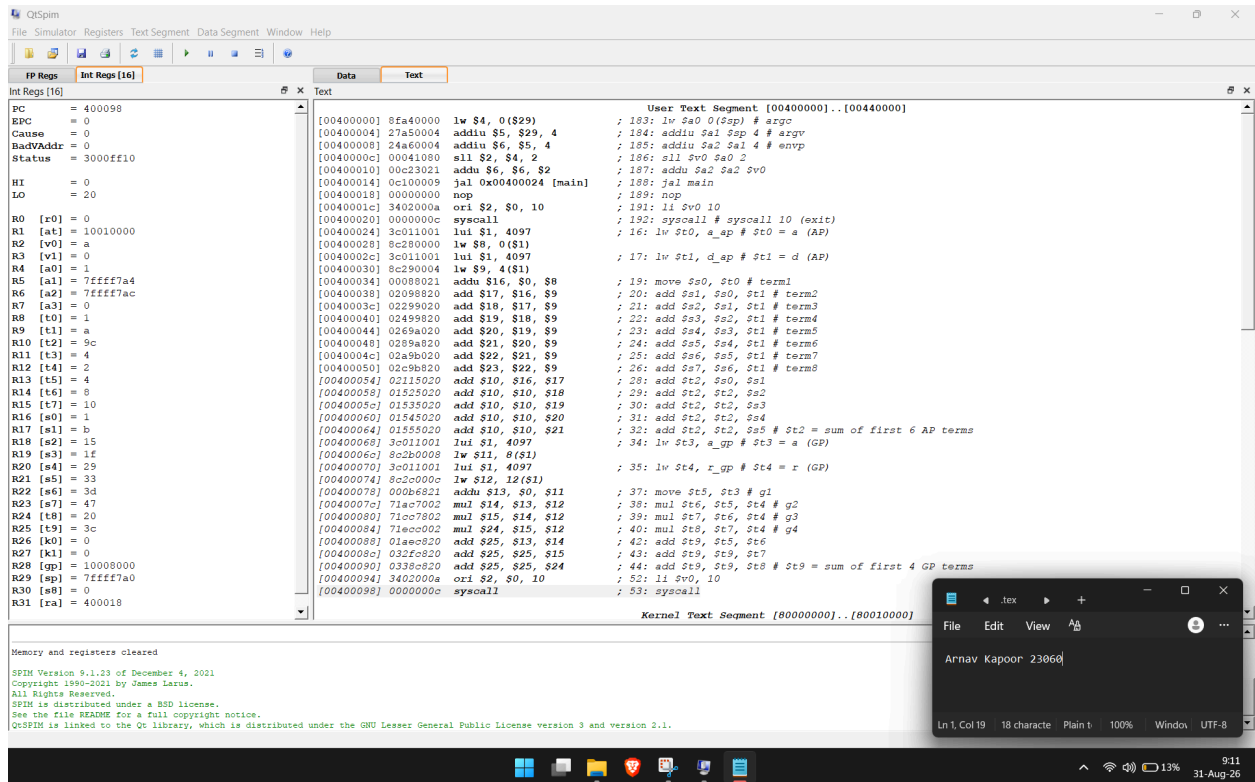
    add  $t2, $s0, $s1
    add  $t2, $t2, $s2
    add  $t2, $t2, $s3
    add  $t2, $t2, $s4
    add  $t2, $t2, $s5

    lw  $t3, a_gp
    lw  $t4, r_gp

    move $t5, $t3
    mul  $t6, $t5, $t4
    mul  $t7, $t6, $t4
    mul  $t8, $t7, $t4
```

```
add $t9, $t5, $t6
add $t9, $t9, $t7
add $t9, $t9, $t8
```

```
li $v0, 10
syscall
```



PROBLEM 2

```
.data
    A_val: .word 1
    B_val: .word 0

.text
.globl main
main:
    lw    $t0, A_val
    lw    $t1, B_val

    # Half Adder
    and   $t2, $t0, $t1    # t2 = A AND B
    or    $t3, $t0, $t1    # t3 = A OR B
    nor   $t4, $t2, $t2    # t4 = NOT(A AND B)
    and   $s0, $t3, $t4    # s0 = SUM = A XOR B
    move  $s1, $t2         # s1 = CARRY = A AND B

    # Half Subtractor
    nor   $t5, $t0, $t0    # t5 = NOT A
    and   $s2, $t5, $t1    # s2 = BORROW = (NOT A) AND B
    move  $s3, $s0         # s3 = DIFFERENCE = A XOR B

    li    $v0, 10
    syscall
```

QtSpim

File Simulator Registers Text Segment Data Segment Window Help

IP Regs Int Regs [16] Data Text

Int Regs [16]

PC = 400058
EPC = 0
Cause = 0
BadVAddr = 0
Status = 3000fff10
HI = 0
LO = 0
R0 [x0] = 0
R1 [at] = 10010000
R2 [v0] = a
R3 [v1] = 0
R4 [a0] = 1
R5 [a1] = 7ffff7a4
R6 [a2] = 7ffff7ac
R7 [a3] = 0
R8 [t0] = 1
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 1
R12 [t4] = ffffffff
R13 [t5] = ffffffff
R14 [t6] = 0
R15 [t7] = 0
R16 [a0] = 1
R17 [a1] = 0
R18 [a2] = 0
R19 [a3] = 1
R20 [a4] = 0
R21 [a5] = 0
R22 [a6] = 0
R23 [a7] = 0
R24 [t8] = 0
R25 [t9] = 0
R26 [k0] = 0
R27 [k1] = 0
R28 [gp] = 10008000
R29 [sp] = 7ffff7a0
R30 [s0] = 0
R31 [ra] = 400018

User Text Segment [00400000]...[00440000]
; 183: lw \$a0 0(\$sp) # argc
; 184: addiu \$a1 \$sp 4 # argv
; 185: addiu \$a2 \$a1 4 # envp
; 186: sll \$v0 \$a0 2
; 187: addu \$a2 \$a2 \$v0
; 188: jal main
; 189: nop
; 191: li \$v0 10
; 192: syscall # syscall 10 (exit)
; 15: lw \$t0, A_val # \$t0 = A
; 16: lw \$t1, B_val # \$t1 = B
; 19: and \$t2, \$t0, \$t1 # \$t2 = A AND B
; 20: or \$t3, \$t0, \$t1 # \$t3 = A OR B
; 21: nor \$t4, \$t2, \$t2 # \$t4 = NOT(A AND B)
; 22: and \$s0, \$t3, \$t4 # \$s0 = SUM = A XOR B
; 23: move \$s1, \$t2 # \$s1 = CARRY = A AND B
; 26: nor \$t5, \$t0, \$t0 # \$t5 = NOT A
; 27: and \$s2, \$t5, \$t1 # \$s2 = BORROW = (NOT A) AND B
; 28: move \$s3, \$s0 # \$s3 = DIFFERENCE = A XOR B (same as SUM)
; 34: li \$v0, 10
; 35: syscall

Kernel Text Segment [80000000]...[80010000]
; 90: move \$k1 \$at # Save \$at
; 92: sw \$v0 \$1 # Not re-entrant and we can't trust \$sp
; 93: sw \$a0 \$2 # But we need to use these registers
; 95: mfc0 \$k0 \$13 # Cause register
; 96: srl \$a0 \$k0 2 # Extract ExCoDe Field
; 97: andi \$a0 \$a0 0x1f
; 101: li \$v0 4 # syscall 4 (print_str)
; 102: la \$a0 __m1_
; 103: syscall
; 105: li \$v0 1 # syscall 1 (print_int)
; 106: srl \$a0 \$k0 2 # Extract ExCoDe Field
; 107: andi \$a0 \$a0 0x1f
; 108: syscall
; 110: li \$v0 4 # syscall 4 (print_str)

Memory and registers cleared

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Arnav Kapoor 23066

9:12 31-Aug-26

PROBLEM 3

```
.data
    val1: .word 25
    val2: .word 40

.text
.globl main
main:
    lw    $t0, val1
    lw    $t1, val2

    # $t0 = $t0 XOR $t1
    and    $t2, $t0, $t1
    or     $t3, $t0, $t1
    nor    $t4, $t2, $t2
    and    $t0, $t3, $t4

    # $t1 = $t0 XOR $t1
    and    $t2, $t0, $t1
    or     $t3, $t0, $t1
    nor    $t4, $t2, $t2
    and    $t1, $t3, $t4

    # $t0 = $t0 XOR $t1
    and    $t2, $t0, $t1
    or     $t3, $t0, $t1
    nor    $t4, $t2, $t2
    and    $t0, $t3, $t4

    li     $v0, 10
    syscall
```

QtSpim

File Simulator Registers Text Segment Data Segment Window Help

IP Regs Int Regs [16] Data Text

Int Regs [16]

PC = 400068
EPC = 0
Cause = 0
BadVAddr = 0
Status = 3000fff10
HI = 0
LO = 0
R0 [x0] = 0
R1 [at] = 10010000
R2 [v0] = a
R3 [v1] = 0
R4 [a0] = 1
R5 [a1] = 7ffff7a4
R6 [a2] = 7ffff7ac
R7 [a3] = 0
R8 [t0] = 28
R9 [t1] = 19
R10 [t2] = 11
R11 [t3] = 39
R12 [t4] = ffffffff
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [a0] = 0
R17 [a1] = 0
R18 [a2] = 0
R19 [a3] = 0
R20 [a4] = 0
R21 [a5] = 0
R22 [a6] = 0
R23 [a7] = 0
R24 [t8] = 0
R25 [t9] = 0
R26 [k0] = 0
R27 [k1] = 0
R28 [gp] = 10008000
R29 [sp] = 7ffff7a0
R30 [s0] = 0
R31 [ra] = 400018

User Text Segment [00400000]...[00440000]
; 183: lw \$a0 0(\$sp) # argc
; 184: addiu \$a1 \$sp 4 # argv
; 185: addiu \$a2 \$a1 4 # envp
; 186: sll \$v0 \$a0 2
; 187: addu \$a2 \$a2 \$v0
; 188: jal main
; 189: nop
; 191: li \$v0 10
; 192: syscall # syscall 10 (exit)
; 14: lw \$t0, val1 # \$t0 = original value1
; 15: lw \$t1, val2 # \$t1 = original value2
; 18: and \$t2, \$t0, \$t1
; 19: or \$t3, \$t0, \$t1
; 20: nor \$t4, \$t2, \$t2
; 21: and \$t0, \$t3, \$t4
; 24: and \$t2, \$t0, \$t1
; 25: or \$t3, \$t0, \$t1
; 26: nor \$t4, \$t2, \$t2
; 27: and \$t1, \$t3, \$t4
; 30: and \$t2, \$t0, \$t1
; 31: or \$t3, \$t0, \$t1
; 32: nor \$t4, \$t2, \$t2
; 33: and \$t0, \$t3, \$t4
; 39: li \$v0, 10
; 40: syscall

Kernel Text Segment [80000000]...[80010000]
; 80: move \$k1 \$at # Save \$at
; 92: sw \$v0 \$1 # Not re-entrant and we can't trust \$sp
; 93: sw \$a0 \$2 # But we need to use these registers
; 95: mfc0 \$k0 \$13 # Cause register
; 96: srl \$a0 \$k0 2 # Extract Exccode Field
; 97: andi \$a0 \$a0 0x1f
; 101: li \$v0 4 # syscall 4 (print_str)
; 102: la \$a0 __ml_
; 103: syscall
; 105: li \$v0 1 # syscall 1 (print_int)

Memory and registers cleared

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File Edit View

Arnav Kapoor 23066

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9:13 31-Aug-26

PROBLEM 4

```
.data
    A_val: .word 800
    B_val: .word 400
    C_val: .word 200
    D_val: .word 100

.text
.globl main
main:
    lw    $t0, A_val
    lw    $t1, B_val
    lw    $t2, C_val
    lw    $t3, D_val

    srl   $t4, $t0, 3
    srl   $t5, $t1, 2
    srl   $t6, $t2, 1
    srl   $t7, $t3, 1

    add   $t8, $t4, $t5
    add   $t8, $t8, $t6
    add   $a0, $t8, $t7

    li    $v0, 17
    syscall
```

QtSpim

File Simulator Registers Text Segment Data Segment Window Help

IP Regs Int Regs [16] Data Text

Int Regs [16]

PC = 400064
EPC = 0
Cause = 0
BadVAddr = 0
Status = 3000fff10
HI = 0
LO = 0
R0 [x0] = 0
R1 [at] = 10010000
R2 [v0] = 11
R3 [v1] = 0
R4 [a0] = 15e
R5 [a1] = 7ffff7a4
R6 [a2] = 7ffff7ac
R7 [a3] = 0
R8 [t0] = 320
R9 [t1] = 190
R10 [t2] = c8
R11 [t3] = 64
R12 [t4] = 64
R13 [t5] = 64
R14 [t6] = 64
R15 [t7] = 32
R16 [a0] = 0
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
R22 [s6] = 0
R23 [s7] = 0
R24 [t8] = 12c
R25 [t9] = 0
R26 [k0] = 0
R27 [k1] = 0
R28 [gp] = 10008000
R29 [sp] = 7ffff7a0
R30 [s0] = 0
R31 [ra] = 400018

User Text Segment [00400000]...[00440000]
; 183: lw \$a0 0(\$sp) # argc
; 184: addiu \$a1 \$sp 4 # argv
; 185: addiu \$a2 \$a1 4 # envp
; 186: sll \$v0 \$a0 2
; 187: addu \$a2 \$a2 \$v0
; 188: jal main
; 189: nop
; 191: li \$v0 10
; 192: syscall # syscall 10 (exit)
; 19: lw \$t0, A_val
; 20: lw \$t1, B_val
; 21: lw \$t2, C_val
; 22: lw \$t3, D_val
; 24: srl \$t4, \$t0, 3 # t4 = 0.125 * A
; 25: srl \$t5, \$t1, 2 # t5 = 0.25 * B
; 26: srl \$t6, \$t2, 1 # t6 = 0.5 * C
; 27: srl \$t7, \$t3, 1 # t7 = 0.5 * D
; 29: add \$t8, \$t4, \$t5
; 30: add \$t8, \$t8, \$t6
; 31: add \$a0, \$t8, \$t7 # \$a0 = weighted average, returned via exit2
; 33: li \$v0, 17
; 34: syscall

Kernel Text Segment [80000000]...[80010000]
; 90: move \$k1 \$a2 # Save \$a2
; 92: sw \$v0 \$1 # Not re-entrant and we can't trust \$sp
; 93: sw \$a0 \$2 # But we need to use these registers
; 95: mfc0 \$k0 \$13 # Cause register
; 96: srl \$a0 \$k0 2 # Extract ExCoDe Field
; 97: andi \$a0 \$a0 0x1f
; 101: li \$v0 4 # syscall 4 (print_str)
; 102: la \$a0 __ml_
; 103: syscall
; 105: li \$v0 1 # syscall 1 (print_int)
; 106: srl \$a0 \$k0 2 # Extract ExCoDe Field

Memory and registers cleared

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Arnav Kapoor 23060

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